

Chapter 14

Graphical Representation of Transitions and Sequences

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Introduction

The analysis of sequences within sociology originally emerged from the biographical perspective on humans lives (Abbott 1983). Since the 1980s, this qualitative approach shifted toward a more quantitative paradigm of life-course analysis (Aisenbrey and Fasang 2010). Both approaches have a longitudinal notion and are focussed on the investigation of causal effects within processes ordered by elapsing life-time. Whereas the qualitative biographical approach was concentrated on generating hypotheses from single individual histories, the quantitative life-course approach uses large datasets with many individuals trying to apply predominantly hypotheses-testing methods (cp. Heinz 2003; George 2009).¹ Sequence analysis, in its recent developmental stage, is a method which lies in between these approaches with its explorative nature and with its use of a large number of cases. Whereas the earlier applications used only a few cases with short sequences because of limited computer capacities, sequence analysis nowadays is able to deal with large case numbers and long sequences. Hence, it can be used to generate hypotheses and as a pre-stage for the application of hypothesis-testing methods.

Most sequences can be seen as discrete, categorical time-series.² In general, sequential information is complex with respect to three dimensions: first, there is a large number of individuals under investigation; second, studies applying sequence analysis tend to have long sequences, i.e., many time points; and third, they have a certain number of categories. This is what makes the visualisation of sequences

¹ One major objective of the life-course approach is the examination of the embedding of aggregate individual life courses into macro-level contexts and in the linkage between life-courses (cp. Elder et al. 2004).

² This is only true for those sequences based on a time-scale mostly used in sociology. But, there are also studies about sequences that have no time dimension, such as the original application of optimal matching (OMA) on distance or DNA strands.

a challenging task. The exploration of sequential data aimed at inductively deriving ideal types requires a heuristic process in which visualization of the data plays a major role. The visual approach to (graphical) information allows for important insights in relevant structures in the data. Hence, the appropriate visualization of sequential information is important not only for the communication between the analyst and others (as a presentation graph) but also for the methodological research process in itself (as an analytic graph): it constitutes a necessary step of the exploration of sequences. Cleveland (1994) argues with respect to graphical representations that this purpose also constitutes a necessary part of all forms of data analyses: “Numerical reduction methods do not retain the information in the data.” In this view, graphical depiction is a source of new insights: “To regularly miss surprises by failing to probe thoroughly with visualization tools is terribly inefficient [...]” (Cleveland 1994, p. 8). The importance of visual inspection of data is not limited to explorative methods but is also relevant for analytical research strategies, which is wonderfully demonstrated for regression diagnostics by the so-called ‘Anscombe’s quartet’ (Anscombe 1973).

A graph contains encoded information which must be decoded correctly by the reader. If the decoding fails—i.e. if information are obscured or misinterpreted—the whole visualisation process failed (Cleveland 1994, p. 1). Consequently, the creation of graphical displays must be conducted very carefully taking into account the kind of information that has to be transmitted as well as the concrete design of a graph.

This chapter is about the display of sequential information with a focus on sociological issues. It presents different kinds of graphical visualisations and discusses their typical advantages and drawbacks. This assessment is mainly based on a model of graphical perception as well as on elementary principles of graph construction and clear understanding developed by William S. Cleveland (1994).

The rest of this chapter is structured as follows: The next section briefly explains Cleveland’s model of graphical perception while concentrating on those principles that are relevant for graphical sequence representation. After that, different kinds of sequence visualisation are discussed, such as sequence index plots, status proportion plots, modal plots, parallel coordinates plots, and transition plots. The main objective is to provide researchers and readers with a deeper understanding of the appropriate application of these graphs.

Cleveland’s Model of Graphical Perception

Based on different kinds of psychological studies and perception tests, Cleveland developed a model of graphical perception that should help to explain why particular graphical realisations are better than others. Accordingly, he compiles a couple of principles that should help researchers enhance the potential of their graphs re-

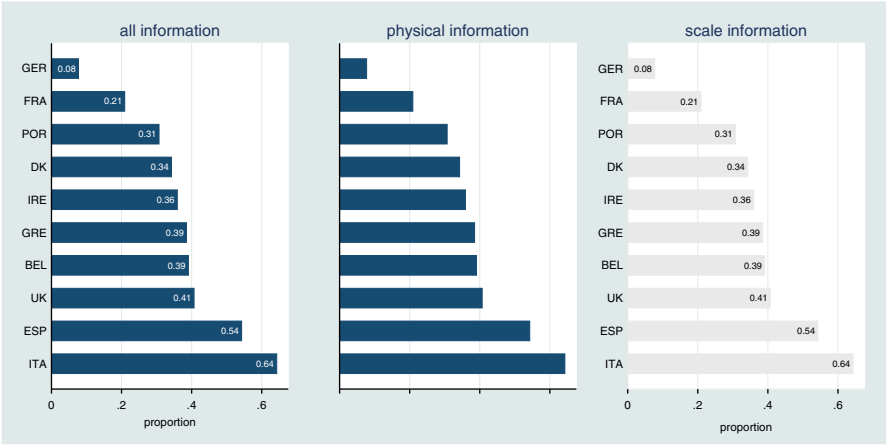


Fig. 14.1 Physical and scale information of a graph

garding information transmission.³ Those principles that are important for sequence visualisation are summarised in this section.

In general, the construction of a graph means encoding information, whereas graphical perception means the decoding of this information. The whole visualisation process fails if the decoding of the information fails (Cleveland 1994, p. 221 f.). All the Information of a graph can be distinguished into scale information and physical information (Fig. 14.1). Scale information is in the name of the category, while physical information is the information that is left when all labels, tick marks, and category names are removed. Physical information refers to the geometry of graphical elements used in a graph.

Scale information is decoded by the so-called ‘table look-up,’ while the visual decoding of physical information is the ‘pattern perception’. For both, efficiency is determined by the speed and accuracy in which the operations are carried out, and fast operations tend to be more accurate. Pattern perception consists of three operations, namely detection, assembly, and estimation. The detection refers to the recognition of the geometric aspects, such as the bars illustrated as having different length, and they are reaching from the left to the right. Assembly is the operation which groups the detected elements together, such as the bars that are showing the same values for different countries. Estimation relates the detected and assembled information by discriminating, ranking, and rationing, such as the longer bars in Fig. 14.1 are at the bottom of the graph. The table look-up also consists of three operations: scanning, interpolation, and matching. Scanning refers to the first recognition of a fix point on the scale, while interpolation describes the estimation of

³ There are many trials of assembling rules for graphical display (Tufte 1983; Wallgren et al. 1996; Blasius and Greenacre 1998; Good and Hardin 2003), but I decided for Cleveland’s principles, because his work is most comprehensive, that is that other rules are included in his aggregation.

the distance between the fix point and a neighbouring tick mark. The matching connects the estimation of the scale distances to the different data points.

These visual operations can be used to gain important information about how graphs are decoded by providing a couple of ideas regarding how to construct them appropriately. Cleveland himself uses them to derive a couple of principles for graphical display and understanding (see Appendix), of which some are of direct relevance for sequence visualisations.⁴ The first principle of graph construction—“Make the data stand out and avoid superfluity” (p. 25)—constitutes already a little problem for the purpose of graphical sequence representation, which is discussed in the respective sections. A second principle is the advice to “use visually prominent graphical elements to show the data” (p. 29). Very important for sequence index plots are the principles that refer to overlapping (p. 50) and to reduction and reproduction (p. 53). The requirement that calls for “superposed data sets [to] be readily visually assembled” (p. 51) is important when comparing sequences of different sub-groups. A basic principle of clear understanding—“Put major conclusions into graphical form” (p. 55)—is, of course, valid for all graphs but constitutes a challenge for explorative procedures. This principle, to a certain extent, contradicts two principles of general strategy, namely that “a large amount of quantitative information can be packed into a small region” (p. 110) and that “graphing data should be an iterative, experimental process” (p. 112). These principles are discussed in the following sections of the respective plots.

Sequence Index Plots

The most intuitive and widely used graphical format for the visualisation of sequential information is the sequence index plot. The three dimensions of sequence data (observations, time points, and categories) are plotted in one single, colourful graph. In its basic version, the horizontal axis represents the time dimension starting point on the left end of the scale and finishing on its right end. The observations are represented on the vertical axis so that each line shows the sequence of one single observation (mostly in reference to individuals). Each point in this coordinate system shows a particular category of a particular observation at a particular time point, while the categories themselves are reflected by a respective color. According to Cleveland (1994, p. 210), there are only two useful applications of colour in graphs, one of which is rendering different categories. Color allows for the quick perception of clear-cut differences in the distinct values of a categorical variable;

⁴ The majority of the principles are already implemented by the most statistical software packages, such as Stata. For example, the principle that the data rectangle should be smaller than the scale-line rectangle (Cleveland 1994, p. 31) is followed by Stata for every graphical command.

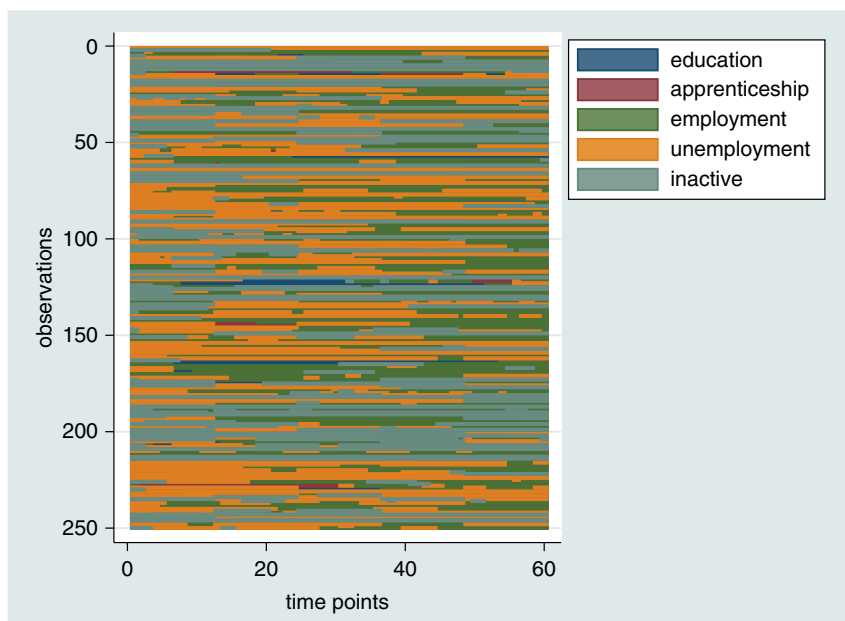


Fig. 14.2 Basic form of sequence index plot

therefore, it is clearly suitable for the indicating different statuses of sequences. An example for a sequence index plot is shown in Fig. 14.2.^{5,6}

The advantage of this graph is its intuitive perception: time is on the horizontal axis, individual sequences have to be read from the left to the right, and the differentiation of the categories is clear because of contrasting colors. In a sequence index plot, the data stand out and they are visually prominent; at the same time, a large amount of information is packed into the graph. Therefore, it serves the purpose to visually detect structures in the bulk of data, and is useful within the iterative, experimental process of sequence exploration. This kind of graph clearly helps to detect unknown characteristics and structures, because “our mind’s eye frequently does not so a good job of predicting what our actual eyes will see” (Cleveland 1994, p. 112). The graph can be modified for the purposes of result communication by changing the colors in a way that similar categories look similar because of lower contrasts.

However, there are also some clear drawbacks of this kind of visualization that arise from the increase in the number of observations or in the number of categories.

⁵ The graphs of this chapter are generated with the software Stata and the user-written sq-commands (Brzinsky-Fay et al. 2006); the data are mainly the same as in Brzinsky-Fay (2007).

⁶ Figures 14.2 and 14.3 show monthly employment status sequences of 250 randomly chosen school leavers from a sample of around 4,000 school leavers in ten European countries. The sequences consist of 60 months (12 years) and are from the European Household Panel (EHP).

A considerable increase in the complexity in one or both of these two dimensions leads to the graph becoming cluttered and not interpretable. This circumstance refers to a violation of the principles in that overlapping symbols must be distinguishable and visual clarity must be preserved under reduction and reproduction. Most of these problems can be solved by modifying the graph in an appropriate manner.

In a sequence index plot, each line represents a chronological sequence of statuses for a given analytical unit. With the increasing number of observations, the lines that have to be displayed in the graph (and whose width is calculated) become thinner. Technically, this constitutes a problem as soon as the calculated line width is smaller than the smallest possible pixel of the graph. In this case, earlier drawn lines are overplotted by later drawn lines.⁷ As a result, some of the sequences are not or not fully visualized and the information in the graph is distorted, because it reflects not the full information anymore. Mainly, there are two strategies to solve this problem of overplotting: (1) reducing the number of observations displayed in the graph, and (2) subtly influencing the order of a sequence.

The first strategy is the reduction of the observation number, which can be achieved in different ways. One can select a random sample from the sequences and display the sequences in a standard sequence index plot. A sequence index plot is able to show between 300 and 400 lines in each graph without overplotting. This strategy clearly inhibits overplotting because with a lower number of cases, the single lines become thicker. However, even a random selection of the sample has a certain likelihood of not reflecting all properties of the original dataset, so that characteristics can be blurred because of sampling. Hence, the sampling should be repeatedly conducted in order to control this effect. Alternatively, the number of observations can be easily reduced by creating one sequence index plot for each category of a particular breakdown variable (see Fig. 14.5). Another possibility is the display of those sequences which are most frequent in the data.⁸ The disadvantage with this variant is that it disregards a lot of information of the data, and it does so systematically by using the frequency as a selection criterion. Whereas in a random sample of the sequences each sequence has the same probability of being displayed in the graph, more frequent sequences have a higher probability. In social science applications of sequence analysis, very often the less frequent sequences are of high interest.

The second strategy refers to the order of the sequences. The negative effect of overplotting can be diminished if the overplotted sequence is equal or similar to the overplotting sequence. This occurs if similar sequences are plotted beside each other. The respective order must be based on a variable that accounts for similarity between the sequences. While in Fig. 14.2 the sequences are displayed in random order, in Fig. 14.3, they are ordered according to the membership in a sequence cluster, which is the result of the application of optimal matching and clustering

⁷ In Stata, the graph command plots the graph category by category, so that categories (statuses) with a higher number overplot those with a lower number.

⁸ This is implemented in the TraMineR package (Gabadinho et al. 2011a; b) for the software R under the name of ‘sequence frequency plots.’

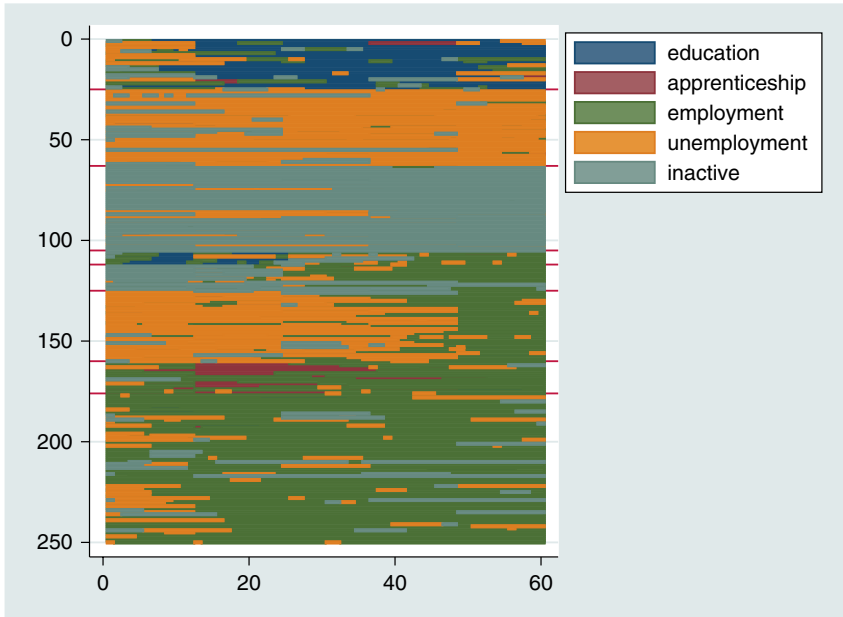


Fig. 14.3 Sequence index plot with ordered sequences

procedure. The observations are the same in both figures, but the clarity regarding the transition types is much higher in Fig. 14.3. Alternatively, the scores of a multi-dimensional scaling may be used to order a sequence index plot (cp. Piccarreta and Lior 2010; Gabadinho et al. 2011a; Fasang and Liao (Forthcoming)).

An interesting combination of reducing the number of observations and changing the order is the so-called representative sequence plot proposed by Gabadinho et al. (2011a). Based on criteria of sequence similarity, the only sequences that are displayed are those representative of multiple sequences from the dataset. Similarly, Piccarreta (2012) provides smoothing techniques, which summarize similar sequences by creating artificial sequences that, while not in the data, are representative of a group of similar sequences. Both techniques solve the problem of overplotting quite well. However, they require pre-conducted comparison procedures and the definition of the selection criteria are quite complex and, to a certain degree, remain arbitrary. Apart from that, the visualized information deviates remarkably from the original data.

The problem of underplotting occurs if a sequence index plot with too few observations is drawn—i.e., if the number of lines (sequences) is much smaller than the number of lines available within a graph. In Fig. 14.4⁹, three sequence index plots are drawn according to the breakdown variable ‘country’. The number of ob-

⁹ Figures 14.4 and 14.5 show monthly employment sequences from school leavers in Belgium, France, and Spain.

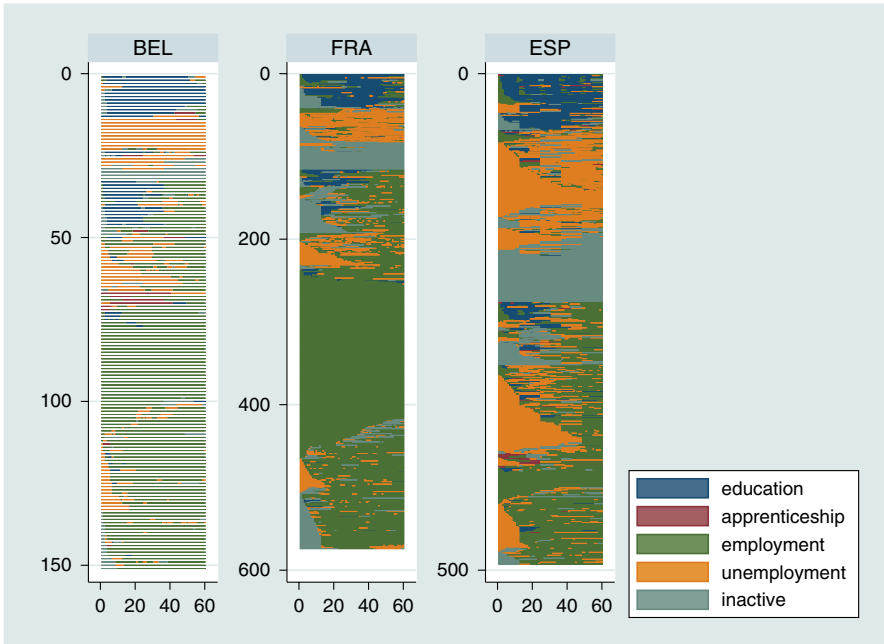


Fig. 14.4 Example for underplotting

servations and the number of lines differ between each of the categories (Belgium, France, Spain). The gaps between the Belgian sequences make the graph nearly unreadable, but they can be changed easily by drawing wide bars instead of thin lines (Fig. 14.5).

Despite customizing the sequence index plot in order to enhance its quality, some of Cleveland's graphing principles remain unsolved: first, only to put major conclusions into graphical form, and second, the preservation of visual clarity under reduction and reproduction. In regard to major conclusions, the violation of the principle stems from the explorative nature of sequence index plots which requires the display of more information than necessary in order to find hidden structures in the data. Apart from this issue, the limitation to major conclusions seems to apply to a larger extent to presentation graphs rather than to graphs serving an iterative research procedure. For the latter case, conclusions have yet to be drawn and the visualisation is needed to find results.

Preserving the visual clarity under print reproduction continues to be a challenging task. The reason can mainly be found in the production and printing process of articles and book chapters (such as the chapter at hand). While the researcher creates graphs on the screen using a particular software, the printing process occurs at a different location in a separate organization using different soft- and hardware. This frequently leads to incompatibilities between different graph formats or at least limits the necessary iteration among constructing the graph, screen view, printing

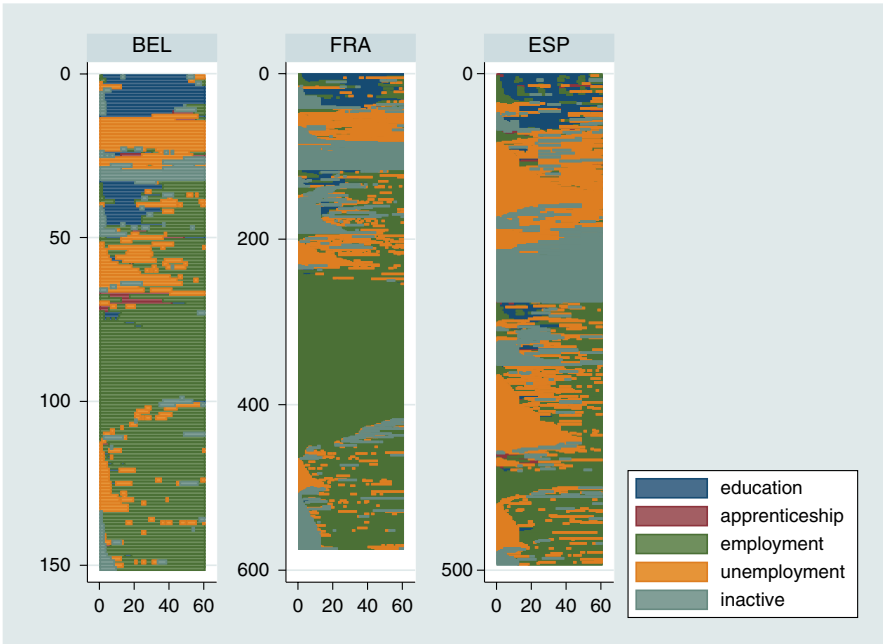


Fig. 14.5 Solution for underplotting

and changing the graph. Technically, it is recommended to produce graphs in either Portable Document Format (.pdf) or PostScript (.ps,.eps) in order to avoid severe damage in the appearance of sequence index plots.

Status Proportion or State Distribution Plots

In reference to one of the principles of clear understanding, namely that only major conclusions should be put into graphical form, the information of a sequence index plot can be condensed for a clearer interpretation of a particular characteristic. A status proportion or state distribution plot displays the relative proportion of each category (status) for every time point. On the horizontal axis, time is displayed, whereas the vertical axis is a percentage scale. Figure 14.6 shows a status proportion plot that uses the same data as Fig. 14.5. It is concentrated only on two of the three dimensions—time and categories—while giving up the observation (individual) dimension. Thus, the construction of a status proportion plot means one step beyond the principle of displaying a large amount of information to the principle of putting only major conclusions into graphical form. The major conclusion here is that in France and Belgium, the share of employment statuses across time is quite similar, whereas in Spain, the monthly share of these statuses differs clearly. The

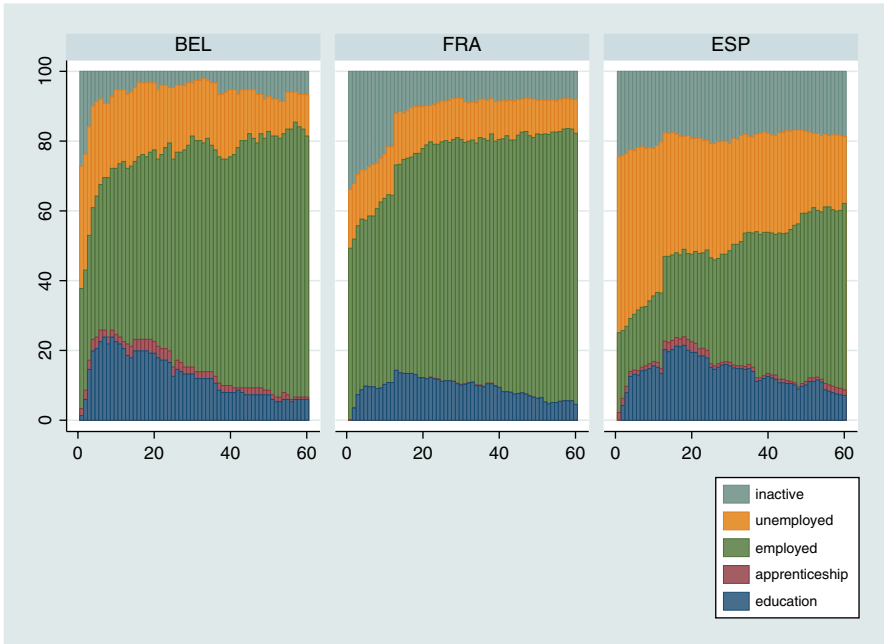


Fig. 14.6 Status proportion plot

reader has to keep in mind that there is no information in this graph concerning individual sequences. The horizontal lines in the graph do not have any meaning in this kind of graph, whereas in the sequence index plot, each horizontal line represents one observation (individual).

Status proportion plots have the advantage of avoiding the violation of the principles of overlapping and preservation of visual clarity under reduction and reproduction. Over- and underplotting does not play a role in this graph. If the colors are chosen in order to provide high contrast¹⁰, changes in size or conversion into black-and-white do not harm the readability. However, this comes at the price of losing the individual information.

Figure 14.7 shows a status proportion plot of the same data used in the other figures but, in this case, sorted differently. Here, the panels of the graph are not showing different countries; rather, they indicate different clusters that are the result of a sequence comparison with optimal matching and a subsequent clustering procedure. The sequence differences between clusters were maximized whereas

¹⁰ In Stata, the standard colors are chosen in order to provide more contrast on color screen and when converted to black-and-white. However, apart from the explorative research procedure being an iterative, experimental process, the choice of the best color scheme is also iterative, and the proof of the pudding is in the eating. For more discussion on the choice of colors for representation of categorical data, see Zeileis et al. (2009).

the differences within clusters were minimized. Therefore, each panel represents a kind of ideal type composed by very similar sequences. Again, the status proportion plot does not display individual sequences; it shows status proportions at each time point: months, in this case. The principle that superposed data sets must be readily and visually assembled is followed by the display of a status proportion plot for each category of the breakdown variable in its own panel.

Modal Plots

A further step into the direction of higher abstraction and information reduction (i.e., showing only major conclusions in the graph) is enabled by constructing modal plots. The modal plot is a summary of the information depicted in a status proportion plot but in a simplified manner. For each time point, the most frequent category (the mode of the distribution) of the sequence is plotted with a horizontal stacked bar chart. In other words, each panel of Fig. 14.7 has now become a horizontal stacked bar in Fig. 14.8. The information of the other categories is discarded because it ignores the variance of categories (statuses) on each time point. But, this graph has the advantage of presenting the clearest possible graphical form of what an ideal type looks like. Bar charts already conform to most of the principles. The principle of putting major conclusions into graphical form is maximized with this kind of graph. Similar to status proportion plots, there are no problems with over- or underplotting, and the visual clarity is maintained under reduction and reproduction, provided that the contrast of the chosen colors survives the conversion into the black-and-white graph.

Sequence index plots, status proportion plots, and modal plots represent different kinds of graphical displays, but they constitute a continuum at the same time. While sequence index plots contain the highest amount of information with a relatively low degree of clarity, modal plots, at the other side of the continuum, contain the lowest amount of information but show the highest clarity. Status proportion plots are located between these two graph types. However, it is not recommended to use only one of these plots because each plot emphasises a focus on different properties of sequential information (Fig. 14.9).

Parallel Coordinate Plots

While the last three graph types are interconnected and are very much part of an integrated explorative research process, parallel coordinate plots constitute a kind of 'stand-alone-display.' The parallel coordinate plot depicts an aggregation of individual sequences and can be used to answer questions for the most frequent se-

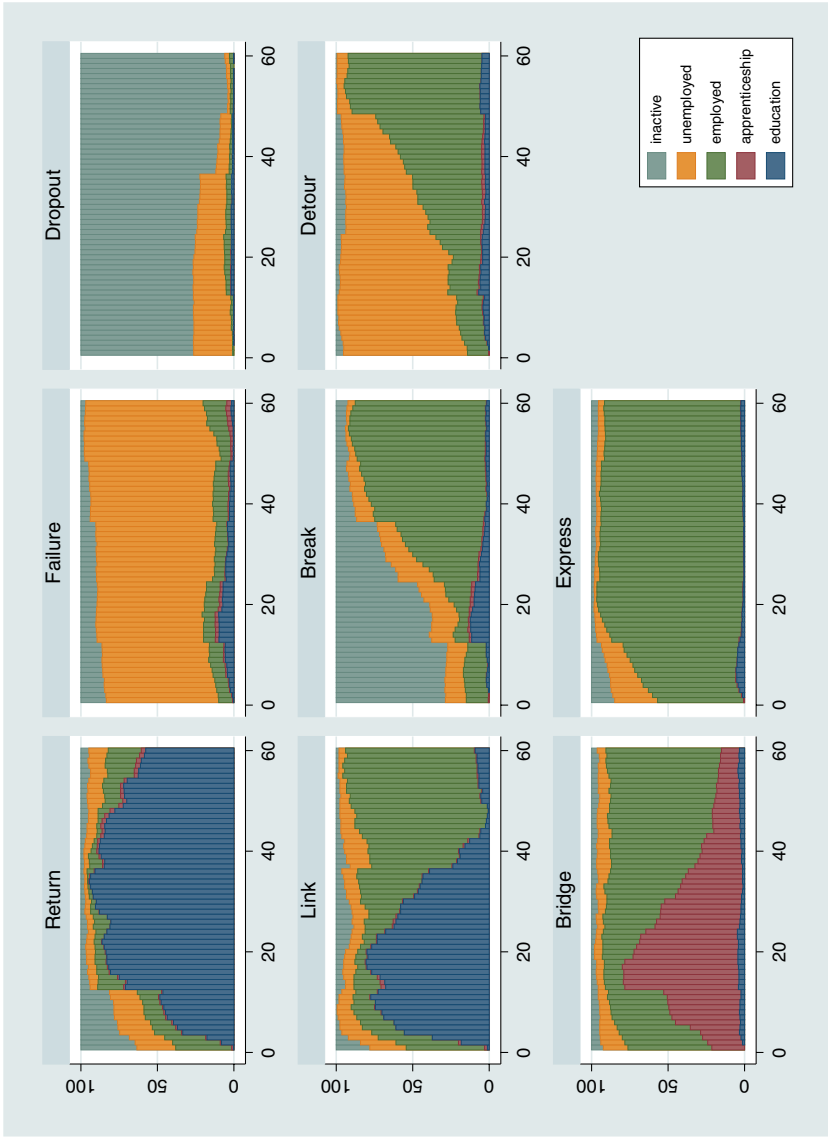


Fig. 14.7 Status proportion plot of sequence types

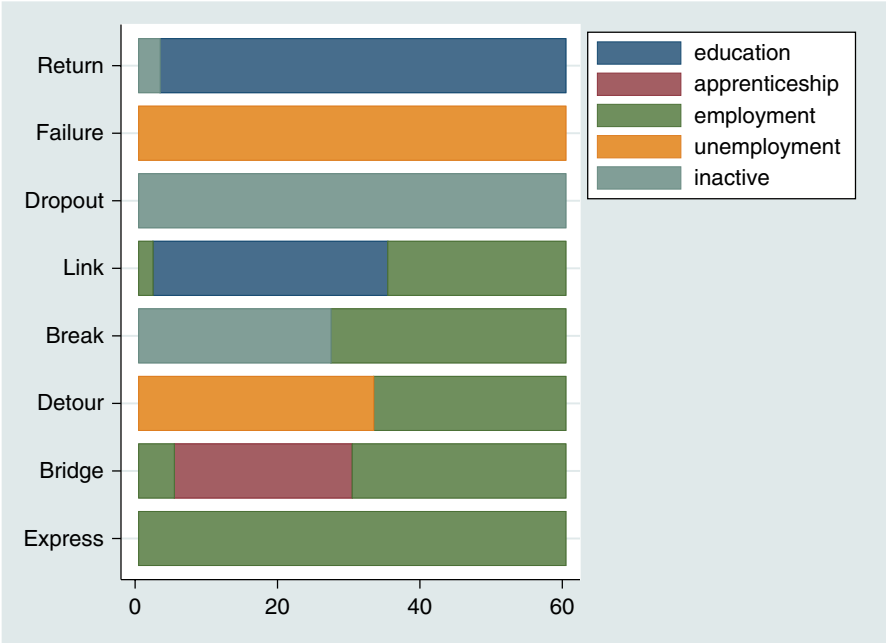
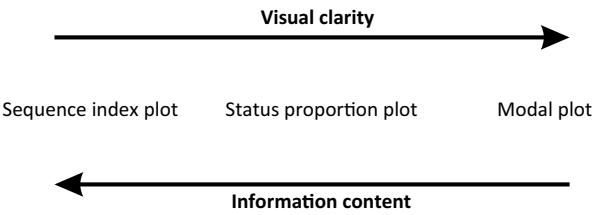


Fig. 14.8 Modal plot of sequence types

Fig. 14.9 Visual clarity and information content of sequence index, status proportion and modal plots



quences. This information can be also gained by simple tables.¹¹ Parallel coordinate plots originally were invented in order to display multidimensional information as an alternative for the Cartesian coordinate system (cp. Theus 2006). They are suitable for the display of sequential information because they can handle more than two dimensions.¹² An example for the data of the Figs. 14.2–14.8 is given in

¹¹ The user-written package TraMineR available for the software R (see footnote 8) offers the possibility to produce so-called sequence frequency plots (Gabadinho et al. 2011a). These plots also show most frequent sequences, and have the form of a weighted bar chart.

¹² However, the original application differs a bit regarding the structure, because each of the vertical axes refers to another dimension, whereas the usage in sequence analysis mixes the Cartesian system with the ‘real’ parallel coordinates in a way that the time points are represented by the horizontal axis.

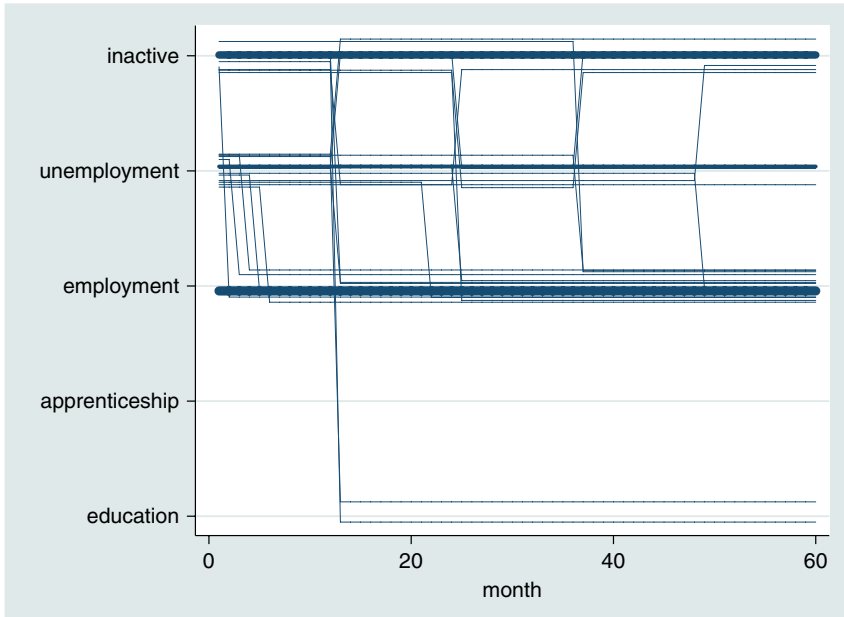


Fig. 14.10 Parallel coordinate plot

Fig. 14.10. On the vertical axis, we find the categories of the sequence states, while the time points are represented by the horizontal axis. The third dimension—the observations—is shown by the lines in the data rectangle. The concession with complexity is the fact that the lines only represent the most frequent sequences in the data set. The number of lines can be defined by the researcher, and their thickness represents the relative frequency. In other words, the thickest lines show the most frequent sequences.

The advantage of this form of graphical display is that the data clearly stand out and the graphical elements (i.e., the lines) are simple and visually prominent. Researchers can easily avoid dispensable information by limiting the display of the lines. Overlapping lines do not constitute a problem because they can be drawn in a jittered way. Major conclusions can be put into graphical form appropriately. Since no colours are necessarily to be used in this graph and the contrasts are very clear, reduction and reproduction don't constitute a problem for visual clarity.

The upper two panels of Fig. 14.11 shows a parallel coordinate plot of the same data, which shows sequence frequencies of two superposed data sets—i.e., two subgroups (males and females). Because the categories are at same vertical position, the principle of easy visual distinction is fulfilled as well. The lower two panels of Fig. 14.11 show an alternative version of this graph, which summarizes the individual sequence data in a slightly different way. The upper two panels show and summarize the sequences with respect to concrete time points, which can be referred to as 'time similarity'. For example, a person A with a sequence that is composed of 10

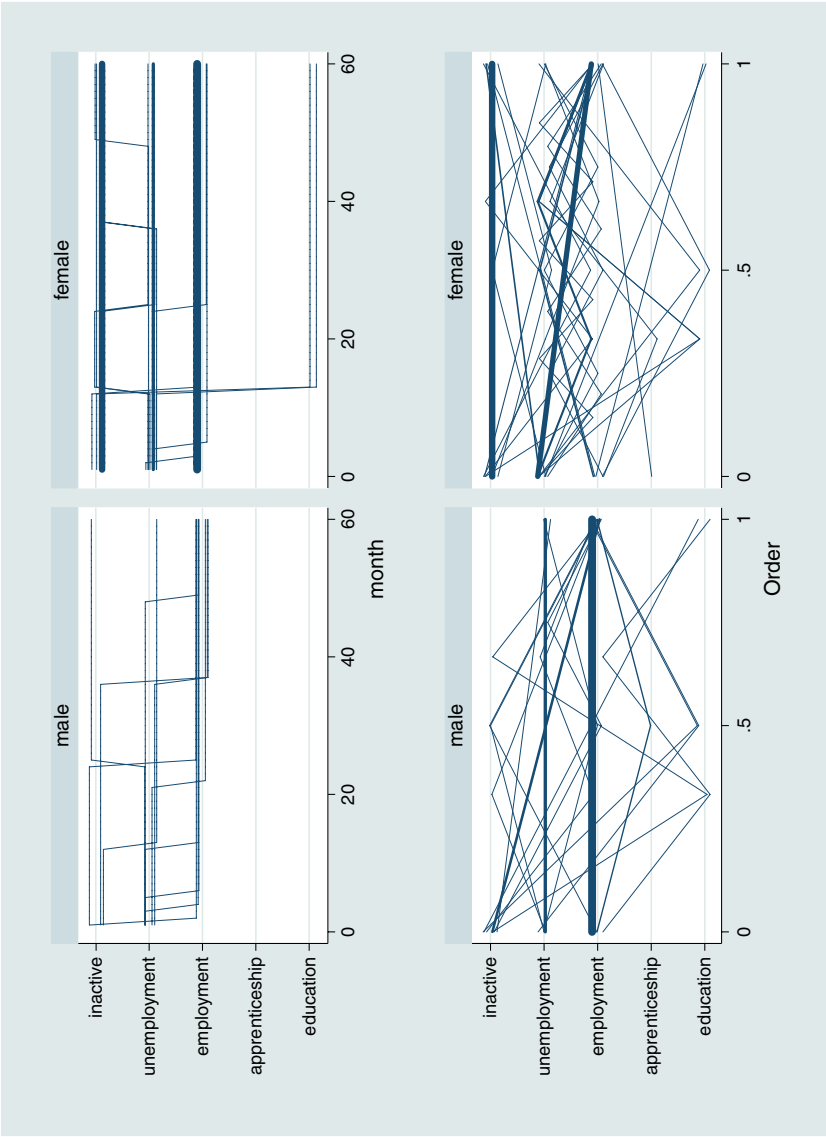


Fig. 14.11 Parallel coordinate plots by gender

months of unemployment followed by 50 months of employment is displayed in a different line than a person B with a sequence that is composed of 20 months of unemployment and 40 months of employment. Both have the same order of statuses, namely unemployment-employment, but because the lengths of the periods differ between person A and person B, they are considered as being different.

In contrast, the parallel coordinate plots in the lower two panels only consider the order of the sequences, while ignoring the absolute length of the episodes. It refers to the so-called ‘same-order similarity’. Here, person A and person B are seen as equal, because the order of statuses is the same. This leads to the result that both sequences are shown in the same line within the parallel coordinate plot, because the similarity refers only to the order—unemployment-employment—which is the same for both persons. Hence, both sequences are summarized in different lines in the upper two panels, while they are summarized in the same line (and constituting its thickness) in the lower two panels.

Regarding the principles of graph construction and clear understanding, there is no difference between the same-order similarity and the time similarity. However, it should be kept in mind that with parallel coordinate plots, there is a substantial loss of information involved because only the most frequent sequences are displayed.

Transition Plots

The transition plot is also helpful for the analysis of sequential information. Transitions are simply changes in the category within the sequence of an observation. They do not genuinely reflect sequential information because they count the number of transition between certain states while disregarding the time dimension and the observation dimension. Figure 14.12 shows a transition plot of the sequence data that also were used for the other figures. The horizontal and the vertical axis show the categories (statuses) of the sequences. Where the reference (dotted) lines do cross, a bubble shows the relative frequency of the transition between the respective statuses from one time point to the other. This graph summarizes all transitions at each time point, but it can be modified so that the transitions for particular time points can be shown. In this graph, the higher the frequency of a transition, the larger the corresponding bubble; hence, the most frequent transition is between employment statuses. The bubbles on the diagonal mean that there is no change or status stability. If one is only interested in changes, then it is helpful not to consider changes between equal statuses—i.e., to not show the bubbles on the diagonal, as it is shown in Fig. 14.13.

The principles are followed in the graphs presented here. Date stand out, and there is no dispensable information in the graph. The graphical elements are visually prominent, overlapping does not exist, and visual clarity is preserved under reduction and (print) reproduction.

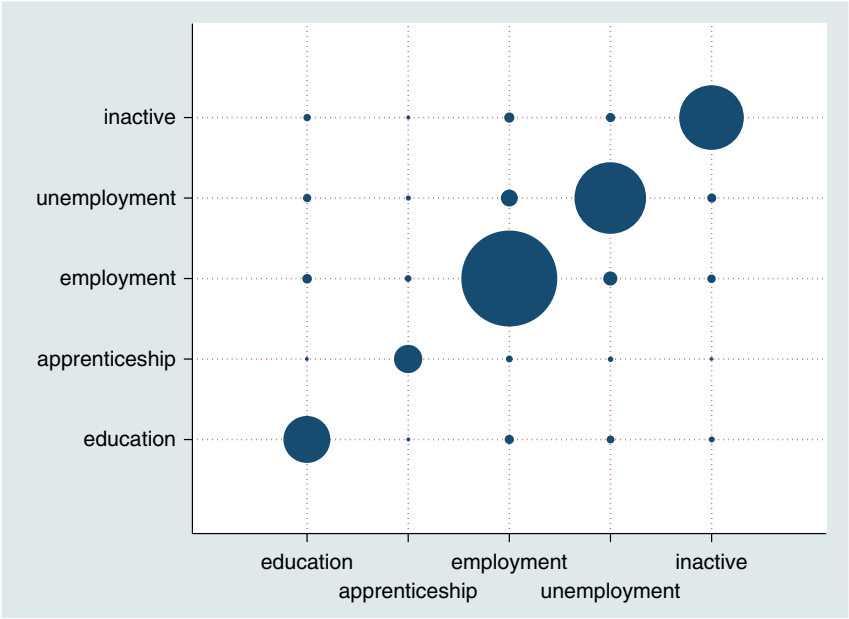


Fig. 14.12 Transition plot with diagonal

Summary

In this chapter, the most useful and applied methods of visualization of sequential information were presented, and their advantages and drawbacks were discussed with respect to principles of graphical display and clear understanding. As categorical time series, sequences contain rich information about individual processes across time, which needs to be presented carefully during the research process of sequence analysis as well as in the end when it comes to result communication. Because of their explorative nature, graphs have a noticeably high relevance even within the research process. Researchers always need to face the trade-off between constructing graphs that are concise in the results they show and that allow for surprising findings regarding hidden structures. The objective of the chapter was to provide sequence analysts insight into graph construction as a means to help them solve this trade-off.

The classical graph for showing results of sequence analyses is still the sequence index plot, predominantly because of its intuitive structure. It fulfils most of the principles of graph construction and clear understanding, but it shows practical problems when it comes to publishing these graphs in printed media. There are some remedies available that can be applied with the statistical software packages able to conduct sequence analysis. Status proportion plots and modal plots are naturally strongly connected to sequence index plots and help to extract important

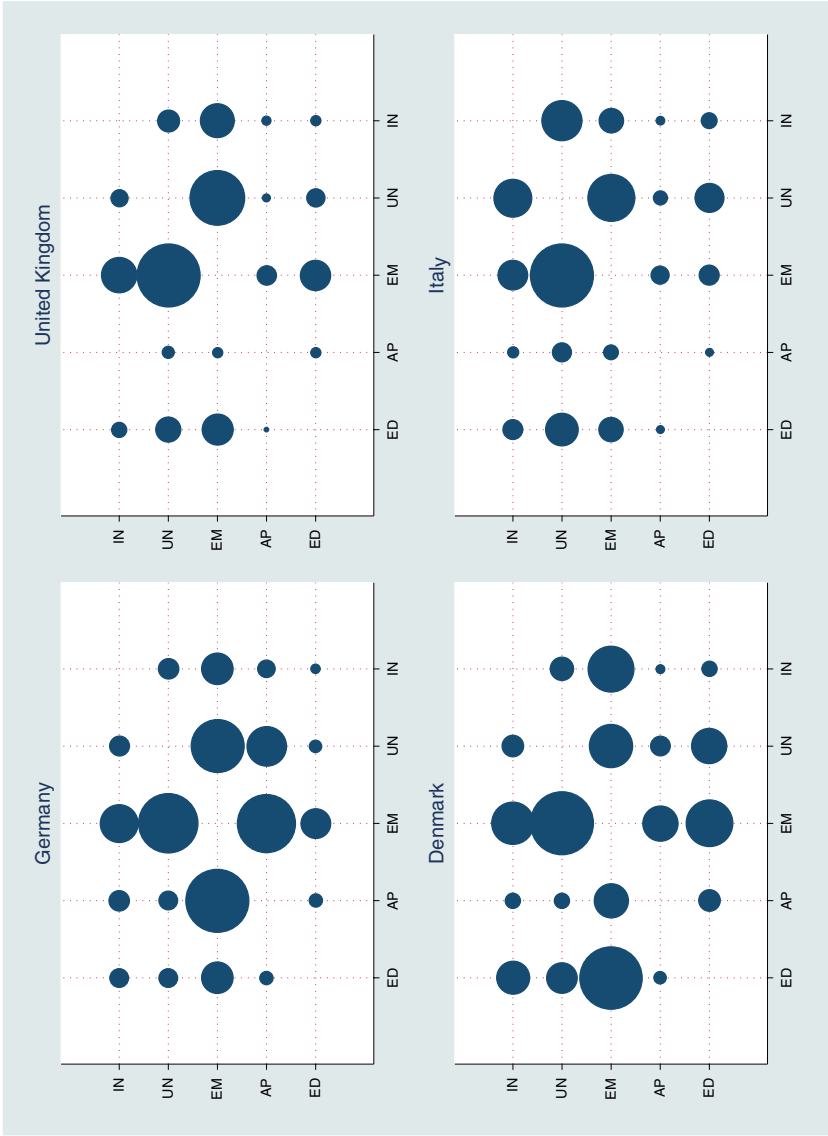


Fig. 14.13 Transition plots by country without diagonal

pieces of information from them. These three graph types constitute a continuum in which the classical trade-off between complexity or information content and visual clarity could be exemplified. The higher the amount of information, the harder it becomes to visualize the results.

Parallel coordinate plots and transition plots are rarely used by sequence analysts, although they provide interesting and valuable information about sequences and transitions. They also follow the principles of graph construction and clear understanding to a larger extent than sequence index plots do. The graphs presented and discussed constitute a rich body of possibilities to successfully conduct an important part of exploratory sequence analysis. Or, as Tukey (1977) explains: They help you “to understand what you CAN DO before you learn to measure how WELL you seem to have DONE it.”

Appendix

This list contains a selection of graphing principles developed and assembled by Cleveland (1994). It is limited to those principles that are relevant for the graphs usually applied to sequence analyses.

Elementary principles of graph construction:

- Make the data stand out & avoid superfluity.
- Use visually prominent graphical elements to show the data.
- [...]
- Overlapping plotting symbols must be visually distinguishable.
- Superposed data sets must be readily visually assembled.
- Visual clarity must be preserved under reduction and reproduction.

Principles of clear understanding:

- Put major conclusions into graphical form.
- [...]

General strategy:

- A large amount of quantitative information can be packed into a small region.
- Graphing data should be an iterative, experimental process.
- [...]

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